



Project Report On Pet-Adoption Database

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A. INTRODUCTION

I. Introduction to Pet Adoption System Database

A Pet Adoption System Database is designed to manage all aspects of an animal adoption center or shelter.

This database stores and organizes data related to pets, adopters, staff members, adoptions, and medical records, ensuring efficient operations, proper record-keeping, and smooth adoption processes.

Key Components:

1. Pets:
 - Stores details about each pet, including name, species, breed, gender..
 - Tracks the availability of pets for adoption.
 - Maintains unique IDs for easy identification.
2. Adopters:
 - Stores adopter information, including name, contact details, address, and adoption history.
 - Enables the shelter to maintain communication with adopters before and after adoption.
3. Staff:
 - Stores staff member details such as name, email, phone number, and role (e.g., Veterinarian, Manager, Caretaker).
 - Helps assign adoption cases and manage daily operations.
4. Adoptions:
 - Records adoption transactions, including date of adoption, adopter information, pet information, and the staff member handling the process.
 - Ensures traceability of all adopted animals.
5. Medical Records:
 - Maintains health and medical details of each pet, including vaccination history, check-up dates, and health status.
 - Helps ensure pets are healthy and ready for adoption.

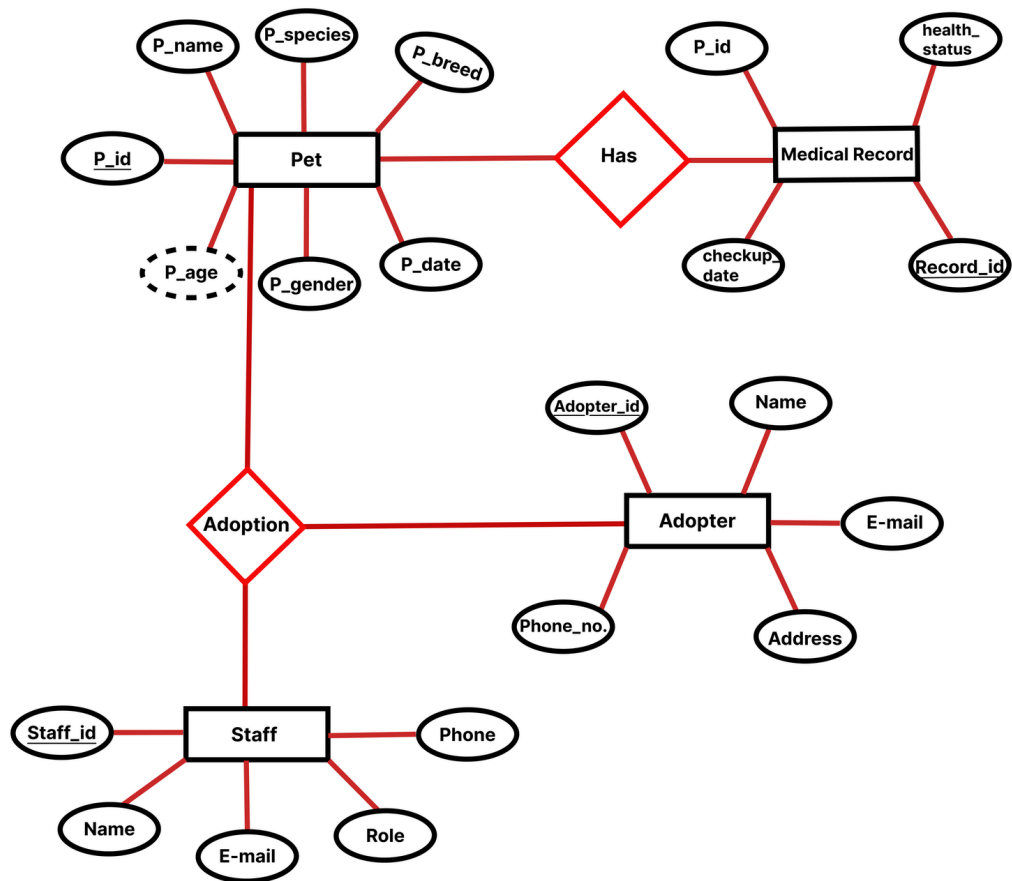
II. Introduction to MySQL

This project uses MySQL 8.1.0, a powerful and widely adopted open-source relational database system. Known for its reliability and scalability, MySQL provides full ACID compliance for safe transactions, advanced high availability through replication and clustering, and robust security features including encryption and fine-grained access control. With enhanced performance optimizations like improved indexing and caching, plus expanded support for JSON and spatial data, MySQL 8.1.0 is ideal for handling diverse and large-scale applications.

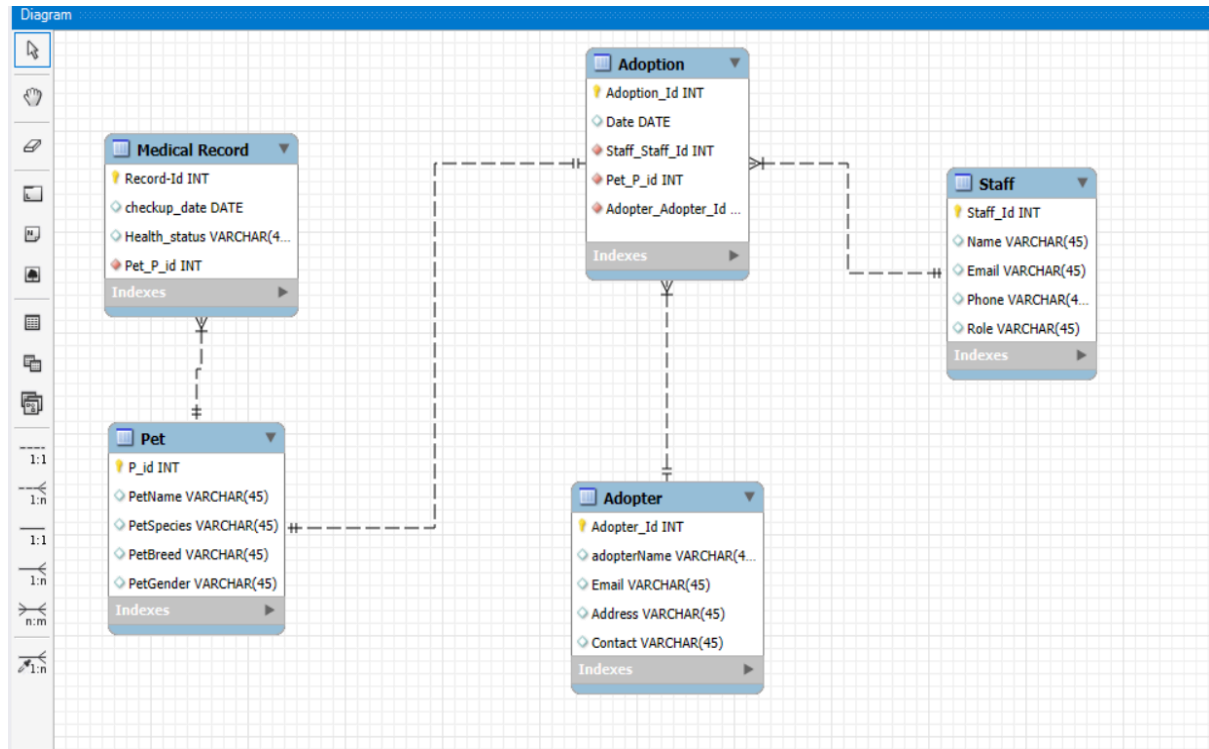
III. Introduction to MySQL Workbench

This project uses MySQL Workbench 8.0.42, a powerful and intuitive visual tool for database design, development, and administration. It features an easy-to-use graphical interface for creating and modeling databases, a rich SQL editor for writing and executing queries, and robust tools for managing server configurations and user accounts. MySQL Workbench also supports seamless data migration from other database systems, comprehensive backup and recovery options, and enhanced performance and stability, making it essential for efficient and reliable database management.

B. ER DIAGRAM



C. DATABASE SCHEMA



D. WORK FLOW

1.1 Pet Adoption Workflow Process

The first step is to evaluate the pet's eligibility for adoption, as follows:

- If the pet has completed its medical checkup and health status is good, it is routed to the next step for adopter verification.
- If the pet's health status is not good or medical checkup is incomplete, it is routed to a Stop node and exits the process.

All eligible pets then go through an adopter verification process.

- If the adopter's information is complete and the adoption request is approved, the system proceeds to create an adoption record.
- If the adopter's information is incomplete or the request is not approved, the workflow stops.

After the adopter is verified, the adoption record must be created, as follows:

- The system creates an entry in the Adoption table with:
- Adopter_Id (who is adopting),
- Pet_P_id (which pet is being adopted),
- Date (adoption date).
- And an unique Adoption_id

After the adoption record is created, the system completes the process as follows:

- The system confirms which staff member handled the adoption and records their Staff_Id in the Adoption table.
- The adoption process then exits the workflow.

E. SQL

1. Database

1.1 CREATE DATABASE PetAdoptionDb

```
CREATE DATABASE PetAdoptionDb;  
USE PetAdoptionDb;
```

```
1 • Create DATABASE PetAdoptionDb;  
2 • USE PetAdoptionDb;
```

Output ::

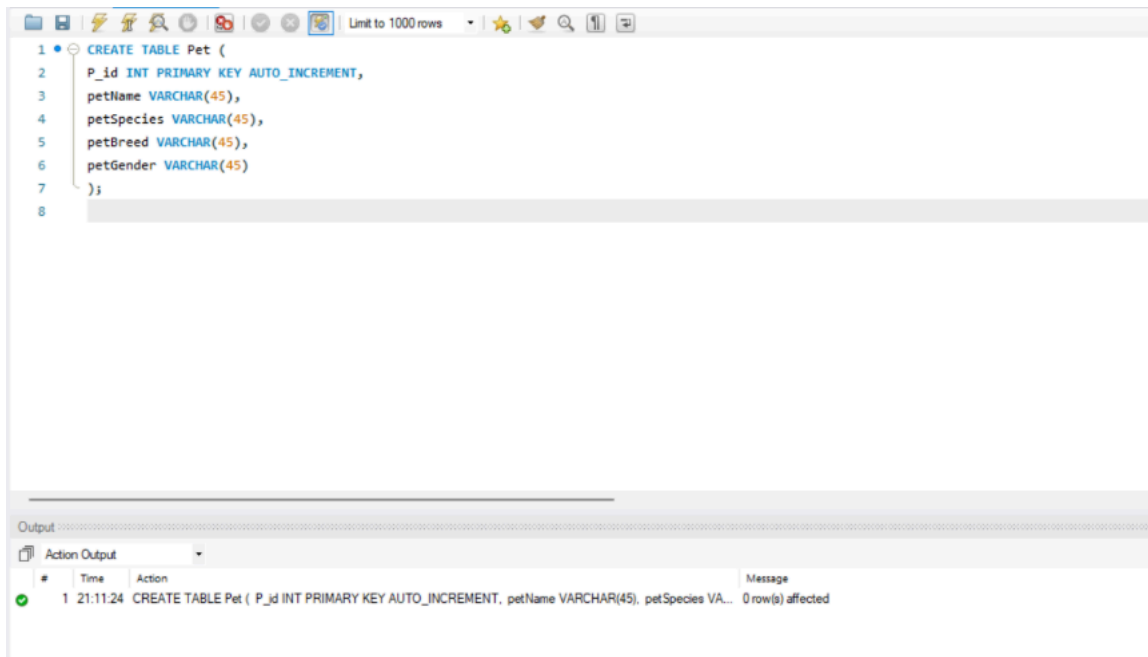
📄 Action Output ▾

	#	Time	Action	Message
✓	1	21:18:20	Create DATABASE PetAdoptionDb	1 row(s) affected
✓	2	21:18:20	USE PetAdoptionDb	0 row(s) affected

2.Tables

2.1 Create Table Pet

```
CREATE TABLE Pet (  
    P_id INT PRIMARY KEY AUTO_INCREMENT,  
    petName VARCHAR(45),  
    petSpecies VARCHAR(45),  
    petBreed VARCHAR(45),  
    petGender VARCHAR(45),  
);
```



2.2 Create Table Adopter

```
CREATE TABLE Adopter (  
    Adopter_Id INT PRIMARY KEY AUTO_INCREMENT,  
    adopterName VARCHAR(45),  
    Email VARCHAR(45),  
    Address VARCHAR(45),
```

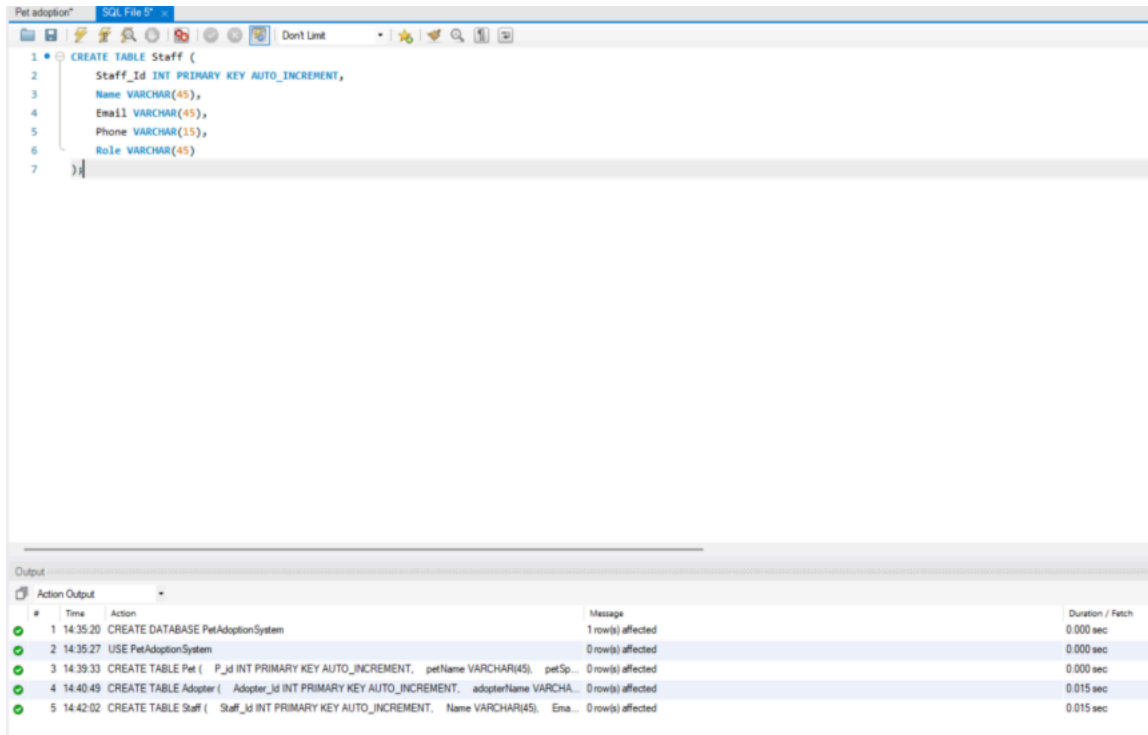
Contact VARCHAR(45),
);

```
1  -- 2. Adopter Table
2  CREATE TABLE Adopter (
3      Adopter_Id INT PRIMARY KEY AUTO_INCREMENT,
4      adopterName VARCHAR(45),
5      Email VARCHAR(45),
6      Address VARCHAR(45),
7      Contact VARCHAR(45)
8  );
9
```

Output			
Action Output			
#	Time	Action	Message
✓ 1	11:42:04	CREATE TABLE Adopter (Adopter_Id INT PRIMARY KEY AUTO_INCREMENT, adopterName VARCHAR(...	0 row(s) affected

2.3 Create Table Staff

```
CREATE TABLE Staff (
    Staff_Id INT PRIMARY KEY AUTO_INCREMENT,
    Name VARCHAR(45),
    Email VARCHAR(45),
    Phone VARCHAR(15),
    Role VARCHAR(45)
);
```

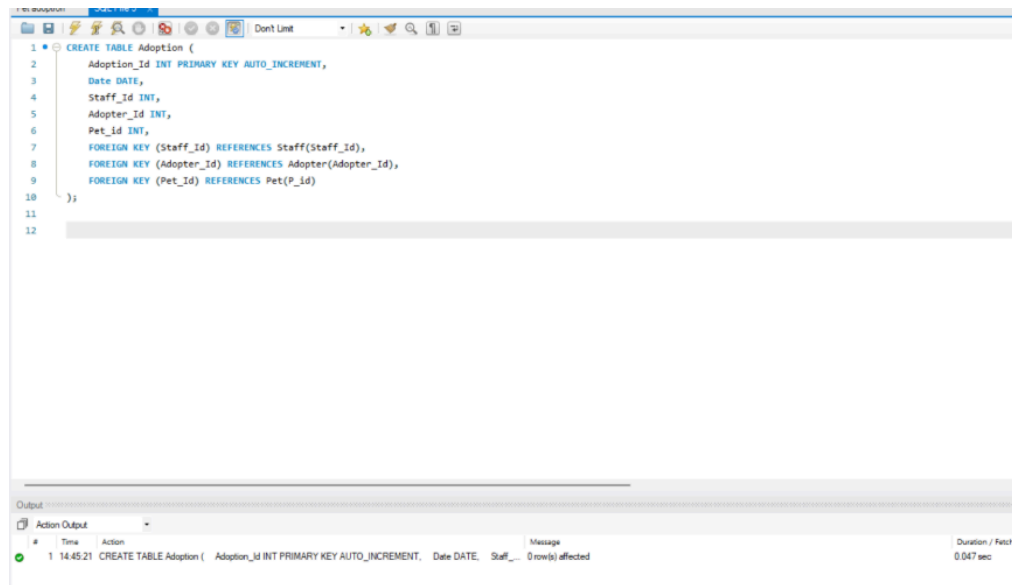


2.4 Create Table Adoption

```

CREATE TABLE Adoption (
    Adoption_Id INT PRIMARY KEY AUTO_INCREMENT,
    Date DATE,
    Staff_Id INT,
    Adopter_Id INT,
    Pet_id INT,
    FOREIGN KEY (Staff_Id) REFERENCES Staff(Staff_Id),
    FOREIGN KEY (Adopter_Id) REFERENCES Adopter(Adopter_Id),
    FOREIGN KEY (Pet_Id) REFERENCES Pet(P_id)
);

```

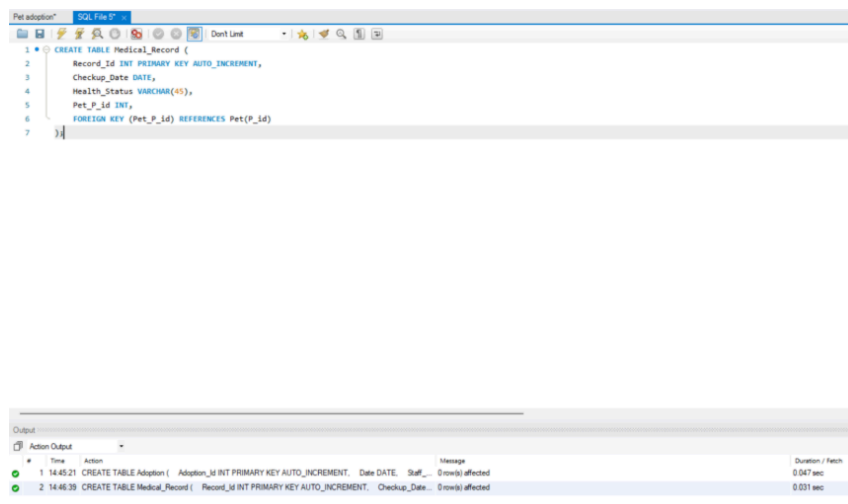


2.5 Create Table Medical_Record

```

CREATE TABLE Medical_Record (
    Record_Id INT PRIMARY KEY AUTO_INCREMENT,
    Checkup_Date DATE,
    Health_Status VARCHAR(45),
    Pet_P_id INT,
    FOREIGN KEY (Pet_P_id) REFERENCES Pet(P_id)
);

```



3. Insert Data

1. Pet table

```
INSERT INTO pet (P_Id, petName, petSpecies, petBreed, petGender) VALUES
(201, 'Moti', 'Dog', 'Nepali Bhote Kukur', 'Male'),
(202, 'Kali', 'Dog', 'German Shepherd', 'Female'),
(203, 'Simba', 'Cat', 'Persian', 'Male'),
(204, 'Gini', 'Cat', 'Local', 'Female'),
(205, 'Bholu', 'Dog', 'Labrador', 'Male'),
(206, 'Luna', 'Cat', 'Siamese', 'Female'),
(207, 'Tiger', 'Dog', 'Mixed', 'Male'),
(208, 'Sweety', 'Cat', 'British Shorthair', 'Female'),
(209, 'Rambo', 'Dog', 'Doberman', 'Male'),
(210, 'Pilu', 'Cat', 'Himalayan', 'Female');
```

A screenshot of a SQL IDE window titled "Pet adoption". The window shows a SQL script in a text editor. The script is an INSERT INTO statement for a table named "pet". The columns are P_Id, petName, petSpecies, petBreed, and petGender. The values are listed in 10 rows, corresponding to the data provided in the text block above. The IDE has a toolbar at the top with various icons for file operations, execution, and search. The script is highlighted in blue and orange colors.

```
1 • INSERT INTO pet (P_Id, petName, petSpecies, petBreed, petGender) VALUES
2   (201, 'Moti', 'Dog', 'Nepali Bhote Kukur', 'Male'),
3   (202, 'Kali', 'Dog', 'German Shepherd', 'Female'),
4   (203, 'Simba', 'Cat', 'Persian', 'Male'),
5   (204, 'Gini', 'Cat', 'Local', 'Female'),
6   (205, 'Bholu', 'Dog', 'Labrador', 'Male'),
7   (206, 'Luna', 'Cat', 'Siamese', 'Female'),
8   (207, 'Tiger', 'Dog', 'Mixed', 'Male'),
9   (208, 'Sweety', 'Cat', 'British Shorthair', 'Female'),
10  (209, 'Rambo', 'Dog', 'Doberman', 'Male'),
11  (210, 'Pilu', 'Cat', 'Himalayan', 'Female');
```

2. Adopter table

```
INSERT INTO adopter (adopter_ID, adopterName, Email, Address, Contact) VALUES
(1, 'Sita Thapa', 'sita.thapa@gmail.com', 'Pokhara, Kaski', '9801234567'),
(2, 'Rajesh Gurung', 'rajesh.gurung@yahoo.com', 'Lalitpur, Nepal', '9812345678'),
(3, 'Maya Shrestha', 'maya.shrestha@hotmail.com', 'Dharan, Sunsari', '9823456789'),
(4, 'Dipesh Bhattarai', 'dipesh.bhattarai@mail.com', 'Butwal, Rupandehi', '9834567890'),
(5, 'Anju Karki', 'anju.karki@gmail.com', 'Bharatpur, Chitwan', '9845678901'),
(6, 'Manoj Lama', 'manoj.lama@gmail.com', 'Hetauda, Makwanpur', '9856789012'),
(7, 'Nisha Adhikari', 'nisha.adhikari@yahoo.com', 'Biratnagar, Morang', '9867890123'),
(8, 'Suraj Poudel', 'suraj.poudel@hotmail.com', 'Damak, Jhapa', '9878901234'),
(9, 'Pratiksha Joshi', 'pratiksha.joshi@mail.com', 'Janakpur, Dhanusha', '9889012345'),
(10, 'Ramesh Basnet', 'ramesh.basnet@gmail.com', 'Nepalgunj, Banke', '9890123456');
```

```
1 • INSERT INTO adopter (adopter_ID, adopterName, Email, Address, Contact) VALUES
2 (1, 'Sita Thapa', 'sita.thapa@gmail.com', 'Pokhara, Kaski', '9801234567'),
3 (2, 'Rajesh Gurung', 'rajesh.gurung@yahoo.com', 'Lalitpur, Nepal', '9812345678'),
4 (3, 'Maya Shrestha', 'maya.shrestha@hotmail.com', 'Dharan, Sunsari', '9823456789'),
5 (4, 'Dipesh Bhattarai', 'dipesh.bhattarai@mail.com', 'Butwal, Rupandehi', '9834567890'),
6 (5, 'Anju Karki', 'anju.karki@gmail.com', 'Bharatpur, Chitwan', '9845678901'),
7 (6, 'Manoj Lama', 'manoj.lama@gmail.com', 'Hetauda, Makwanpur', '9856789012'),
8 (7, 'Nisha Adhikari', 'nisha.adhikari@yahoo.com', 'Biratnagar, Morang', '9867890123'),
9 (8, 'Suraj Poudel', 'suraj.poudel@hotmail.com', 'Damak, Jhapa', '9878901234'),
10 (9, 'Pratiksha Joshi', 'pratiksha.joshi@mail.com', 'Janakpur, Dhanusha', '9889012345'),
11 (10, 'Ramesh Basnet', 'ramesh.basnet@gmail.com', 'Nepalgunj, Banke', '9890123456');
12
```

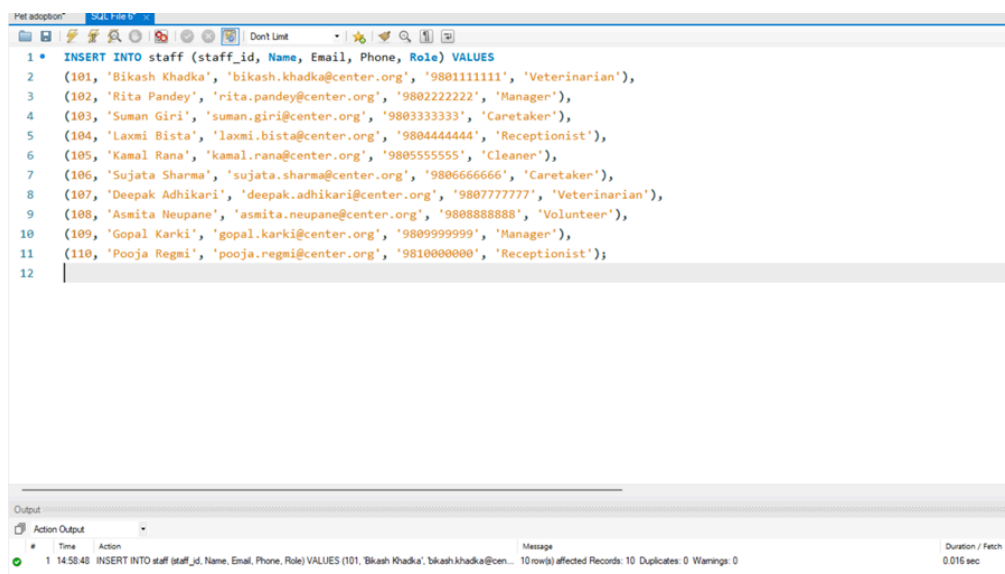
Output

Action Output

#	Time	Action	Message
1	11:43:43	INSERT INTO adopter (adopter_ID, adopterName, Email, Address, Contact) VALUES (1, 'Sita Thapa', 'sita.thapa...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

3. Staff table

```
INSERT INTO staff (staff_id, Name, Email, Phone, Role) VALUES
(101, 'Bikash Khadka', 'bikash.khadka@center.org', '9801111111', 'Veterinarian'),
(102, 'Rita Pandey', 'rita.pandey@center.org', '9802222222', 'Manager'),
(103, 'Suman Giri', 'suman.giri@center.org', '9803333333', 'Caretaker'),
(104, 'Laxmi Bista', 'laxmi.bista@center.org', '9804444444', 'Receptionist'),
(105, 'Kamal Rana', 'kamal.rana@center.org', '9805555555', 'Cleaner'),
(106, 'Sujata Sharma', 'sujata.sharma@center.org', '9806666666', 'Caretaker'),
(107, 'Deepak Adhikari', 'deepak.adhikari@center.org', '9807777777', 'Veterinarian'),
(108, 'Asmita Neupane', 'asmita.neupane@center.org', '9808888888', 'Volunteer'),
(109, 'Gopal Karki', 'gopal.karki@center.org', '9809999999', 'Manager'),
(110, 'Pooja Regmi', 'pooja.regmi@center.org', '9810000000', 'Receptionist');
```



The screenshot shows a SQL Developer window titled 'Pet adoption' with a 'SQL Fiddle' tab. The main editor contains an SQL statement to insert 10 rows into the 'staff' table. The statement is as follows:

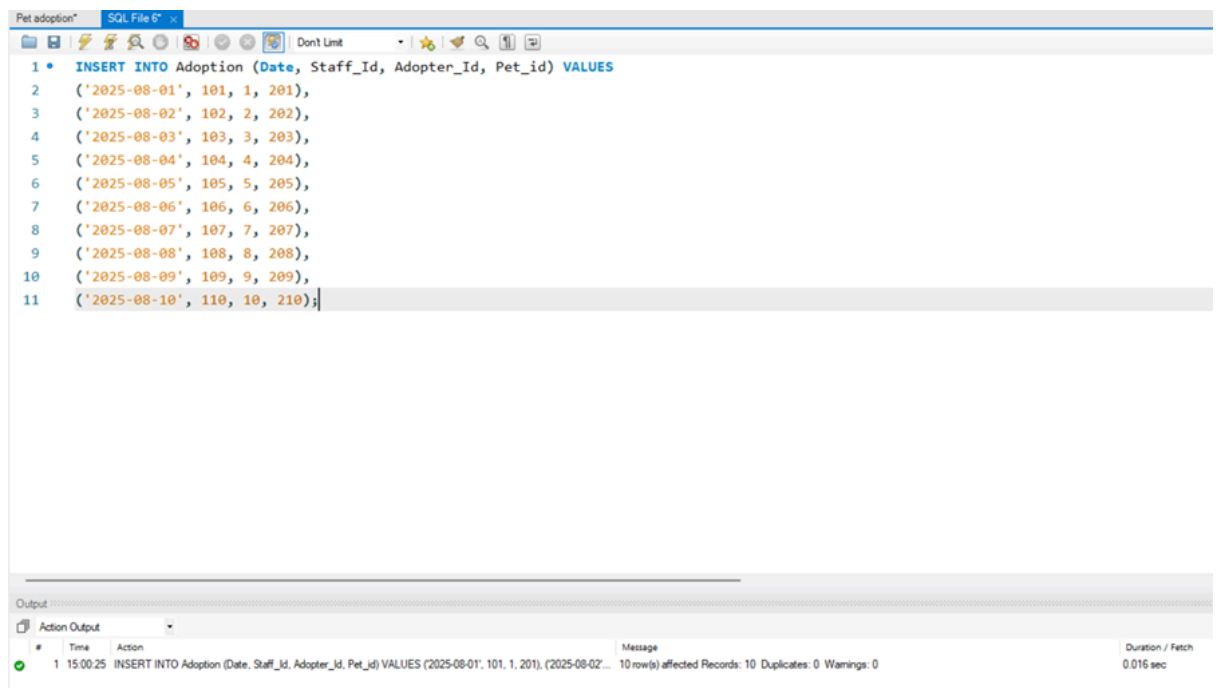
```
1 • INSERT INTO staff (staff_id, Name, Email, Phone, Role) VALUES
2 (101, 'Bikash Khadka', 'bikash.khadka@center.org', '9801111111', 'Veterinarian'),
3 (102, 'Rita Pandey', 'rita.pandey@center.org', '9802222222', 'Manager'),
4 (103, 'Suman Giri', 'suman.giri@center.org', '9803333333', 'Caretaker'),
5 (104, 'Laxmi Bista', 'laxmi.bista@center.org', '9804444444', 'Receptionist'),
6 (105, 'Kamal Rana', 'kamal.rana@center.org', '9805555555', 'Cleaner'),
7 (106, 'Sujata Sharma', 'sujata.sharma@center.org', '9806666666', 'Caretaker'),
8 (107, 'Deepak Adhikari', 'deepak.adhikari@center.org', '9807777777', 'Veterinarian'),
9 (108, 'Asmita Neupane', 'asmita.neupane@center.org', '9808888888', 'Volunteer'),
10 (109, 'Gopal Karki', 'gopal.karki@center.org', '9809999999', 'Manager'),
11 (110, 'Pooja Regmi', 'pooja.regmi@center.org', '9810000000', 'Receptionist');
12
```

Below the editor is an 'Output' window showing the execution results. It displays a single row of data:

#	Time	Action	Message	Duration / Fetch
1	14:58:48	INSERT INTO staff (staff_id, Name, Email, Phone, Role) VALUES (101, 'Bikash Khadka', 'bikash.khadka@center.org', '9801111111', 'Veterinarian'), (102, 'Rita Pandey', 'rita.pandey@center.org', '9802222222', 'Manager'), (103, 'Suman Giri', 'suman.giri@center.org', '9803333333', 'Caretaker'), (104, 'Laxmi Bista', 'laxmi.bista@center.org', '9804444444', 'Receptionist'), (105, 'Kamal Rana', 'kamal.rana@center.org', '9805555555', 'Cleaner'), (106, 'Sujata Sharma', 'sujata.sharma@center.org', '9806666666', 'Caretaker'), (107, 'Deepak Adhikari', 'deepak.adhikari@center.org', '9807777777', 'Veterinarian'), (108, 'Asmita Neupane', 'asmita.neupane@center.org', '9808888888', 'Volunteer'), (109, 'Gopal Karki', 'gopal.karki@center.org', '9809999999', 'Manager'), (110, 'Pooja Regmi', 'pooja.regmi@center.org', '9810000000', 'Receptionist');	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec

4. Adoption table

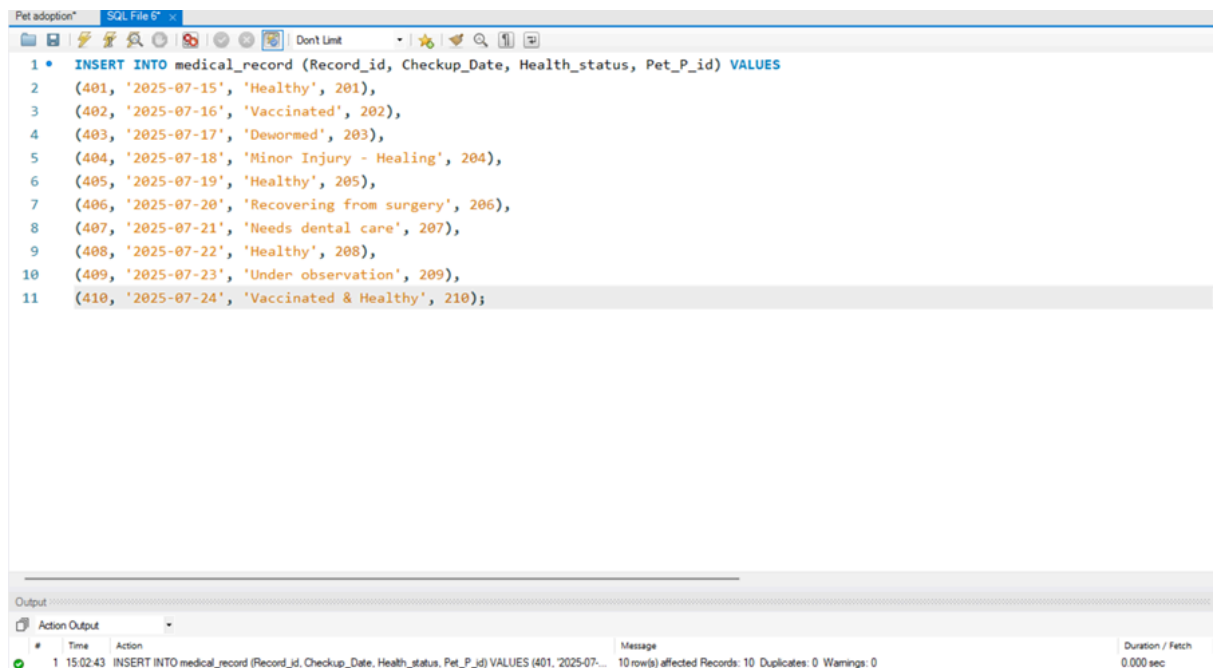
```
INSERT INTO Adoption (Date, Staff_Id, Adopter_Id, Pet_id) VALUES
('2025-08-01', 101, 1, 201),
('2025-08-02', 102, 2, 202),
('2025-08-03', 103, 3, 203),
('2025-08-04', 104, 4, 204),
('2025-08-05', 105, 5, 205),
('2025-08-06', 106, 6, 206),
('2025-08-07', 107, 7, 207),
('2025-08-08', 108, 8, 208),
('2025-08-09', 109, 9, 209),
('2025-08-10', 110, 10, 210);
```



5. Medical_Record table

```
INSERT INTO medical_record (Record_id, Checkup_Date, Health_status, Pet_P_id)
VALUES
```

```
(401, '2025-07-15', 'Healthy', 201),
(402, '2025-07-16', 'Vaccinated', 202),
(403, '2025-07-17', 'Dewormed', 203),
(404, '2025-07-18', 'Minor Injury - Healing', 204),
(405, '2025-07-19', 'Healthy', 205),
(406, '2025-07-20', 'Recovering from surgery', 206),
(407, '2025-07-21', 'Needs dental care', 207),
(408, '2025-07-22', 'Healthy', 208),
(409, '2025-07-23', 'Under observation', 209),
(410, '2025-07-24', 'Vaccinated & Healthy', 210);
```



The screenshot shows a SQL IDE window titled "Pet adoption" with a "SQL File 0" tab. The main editor displays an SQL INSERT statement with 11 lines. The statement inserts 10 records into the "medical_record" table, with columns "Record_id", "Checkup_Date", "Health_status", and "Pet_P_id". The records are numbered 401 to 410. The "Health_status" values include "Healthy", "Vaccinated", "Dewormed", "Minor Injury - Healing", "Recovering from surgery", "Needs dental care", "Under observation", and "Vaccinated & Healthy". The "Pet_P_id" values range from 201 to 210. The bottom panel shows the "Output" tab with a table of execution results. The first row indicates that 10 rows were affected, with no duplicates or warnings, and the operation took 0.000 seconds.

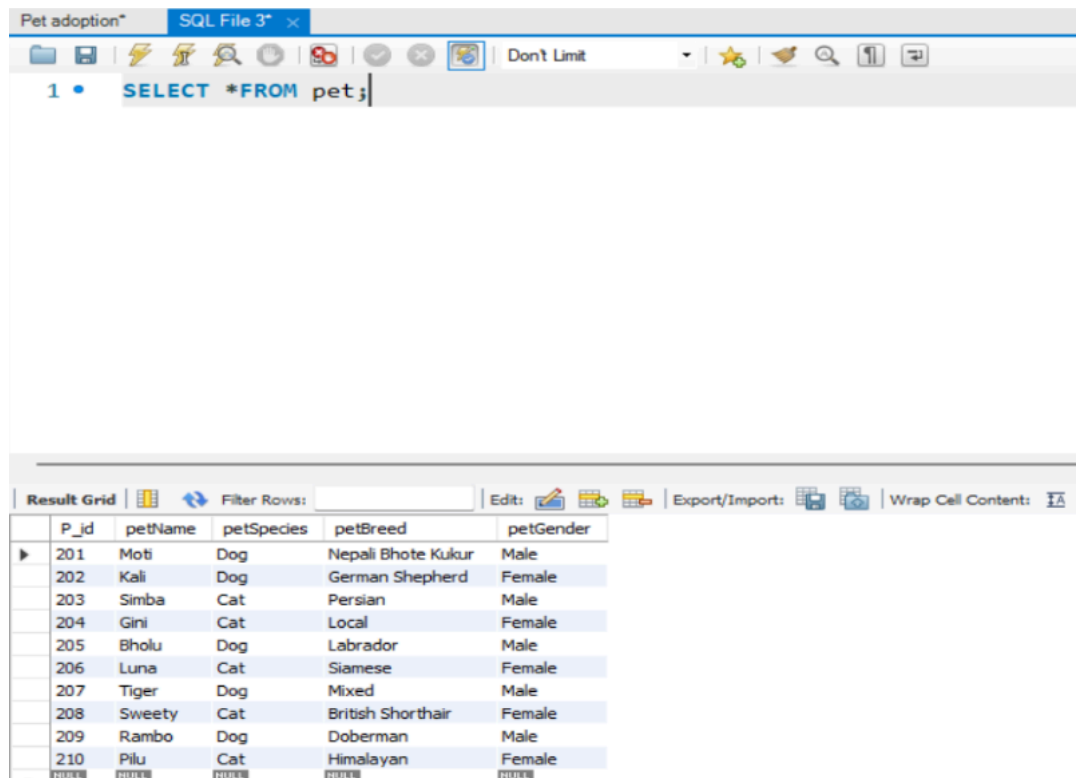
```
1 • INSERT INTO medical_record (Record_id, Checkup_Date, Health_status, Pet_P_id) VALUES
2 (401, '2025-07-15', 'Healthy', 201),
3 (402, '2025-07-16', 'Vaccinated', 202),
4 (403, '2025-07-17', 'Dewormed', 203),
5 (404, '2025-07-18', 'Minor Injury - Healing', 204),
6 (405, '2025-07-19', 'Healthy', 205),
7 (406, '2025-07-20', 'Recovering from surgery', 206),
8 (407, '2025-07-21', 'Needs dental care', 207),
9 (408, '2025-07-22', 'Healthy', 208),
10 (409, '2025-07-23', 'Under observation', 209),
11 (410, '2025-07-24', 'Vaccinated & Healthy', 210);
```

#	Time	Action	Message	Duration / Fetch
1	15:02:43	INSERT INTO medical_record (Record_id, Checkup_Date, Health_status, Pet_P_id) VALUES (401, '2025-07-15', 'Healthy', 201), (402, '2025-07-16', 'Vaccinated', 202), (403, '2025-07-17', 'Dewormed', 203), (404, '2025-07-18', 'Minor Injury - Healing', 204), (405, '2025-07-19', 'Healthy', 205), (406, '2025-07-20', 'Recovering from surgery', 206), (407, '2025-07-21', 'Needs dental care', 207), (408, '2025-07-22', 'Healthy', 208), (409, '2025-07-23', 'Under observation', 209), (410, '2025-07-24', 'Vaccinated & Healthy', 210);	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec

4. Retrieve all the records from each table

I. Retrieve all record from Pet

SELECT *FROM pet;



The screenshot shows a SQL IDE window titled 'Pet adoption*' with a tab for 'SQL File 3'. The query editor contains the SQL statement 'SELECT *FROM pet;'. Below the editor, the 'Result Grid' displays the data from the 'pet' table. The grid has columns for P_id, petName, petSpecies, petBreed, and petGender. The data is as follows:

P_id	petName	petSpecies	petBreed	petGender
201	Moti	Dog	Nepali Bhote Kukur	Male
202	Kali	Dog	German Shepherd	Female
203	Simba	Cat	Persian	Male
204	Gini	Cat	Local	Female
205	Bholu	Dog	Labrador	Male
206	Luna	Cat	Siamese	Female
207	Tiger	Dog	Mixed	Male
208	Sweetie	Cat	British Shorthair	Female
209	Rambo	Dog	Doberman	Male
210	Pilu	Cat	Himalayan	Female
	NULL	NULL	NULL	NULL

II. Retrieve all record from Adopter

SELECT *FROM adopter;

SQL File 10* SQL File 13* x

Limit to 1000 rows

1 • `SELECT *FROM adopter;`

Result Grid

Adopter_Id	adopterName	Email	Address	Contact
1	Sita Thapa	sita.thapa@gmail.com	Pokhara, Kaski	9801234567
2	Rajesh Gurung	rajesh.gurung@yahoo.com	Lalitpur, Nepal	9812345678
3	Maya Shrestha	maya.shrestha@hotmail.com	Dharan, Sunsari	9823456789
4	Dipesh Bhattarai	dipesh.bhattarai@gmail.com	Butwal, Rupandehi	9834567890
5	Anju Karki	anju.karki@gmail.com	Bharatpur, Chitwan	9845678901
6	Manoj Lama	manoj.lama@gmail.com	Hetauda, Makwanpur	9856789012
7	Nisha Adhikari	nisha.adhikari@yahoo.com	Biratnagar, Morang	9867890123
8	Suraj Poudel	suraj.poudel@hotmail.com	Damak, Jhapa	9878901234
9	Pratiksha Joshi	pratiksha.joshi@gmail.com	Janakpur, Dhanusha	9889012345
10	Ramesh Basnet	ramesh.basnet@gmail.com	Nepalgunj, Barke	9890123456
NULL	NULL	NULL	NULL	NULL

adopter 1 x

Output

Action Output

#	Time	Action	Message
1	11:44:48	SELECT *FROM adopter LIMIT 0. 1000	10 row(s) returned

III. Retrieve all record from Staff

`SELECT *FROM staff;`

Pet adoption* SQL File 7* x

Don't Limit

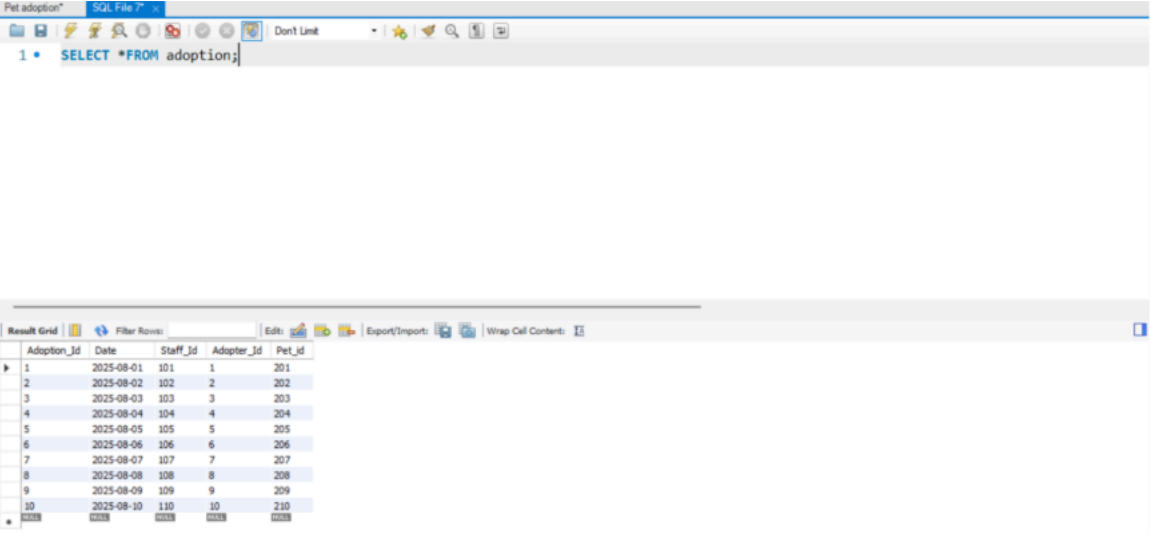
1 • `SELECT *FROM staff;`

Result Grid

Staff_Id	Name	Email	Phone	Role
101	Bikash Khadka	bikash.khadka@center.org	9801111111	Veterinarian
102	Rita Pandey	rita.pandey@center.org	9802222222	Manager
103	Suman Giri	suman.giri@center.org	9803333333	Caretaker
104	Laxmi Bista	laxmi.bista@center.org	9804444444	Receptionist
105	Kamal Rana	kamal.rana@center.org	9805555555	Cleaner
106	Sujata Sharma	sujata.sharma@center.org	9806666666	Caretaker
107	Deepak Adhikari	deepak.adhikari@center.org	9807777777	Veterinarian
108	Asmita Neupane	asmita.neupane@center.org	9808888888	Volunteer
109	Gopal Karki	gopal.karki@center.org	9809999999	Manager
110	Pooja Regmi	pooja.regmi@center.org	9810000000	Receptionist
NULL	NULL	NULL	NULL	NULL

IV. Retrieve all record from Adoption

SELECT *FROM adoption;

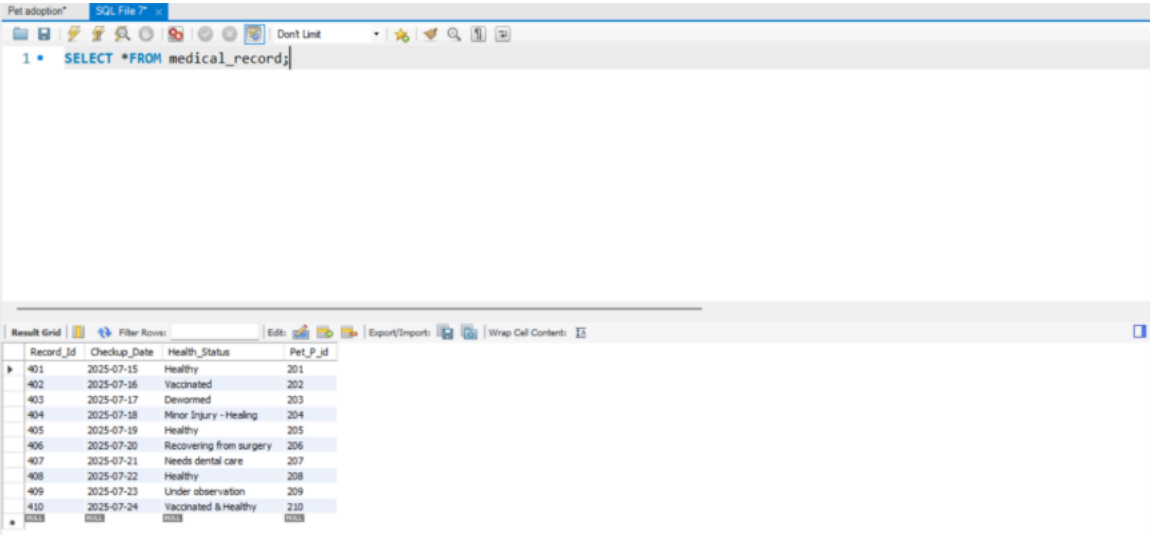


The screenshot shows a SQL File 7 window with the query `SELECT *FROM adoption;` entered. Below the query editor, the 'Result Grid' displays the following data:

Adoption_Id	Date	Staff_Id	Adopter_Id	Pet_Id
1	2025-08-01	101	1	201
2	2025-08-02	102	2	202
3	2025-08-03	103	3	203
4	2025-08-04	104	4	204
5	2025-08-05	105	5	205
6	2025-08-06	106	6	206
7	2025-08-07	107	7	207
8	2025-08-08	108	8	208
9	2025-08-09	109	9	209
10	2025-08-10	110	10	210

V. Retrieve all record from Medical_Record

SELECT *FROM medical_record;



The screenshot shows a SQL File 7 window with the query `SELECT *FROM medical_record;` entered. Below the query editor, the 'Result Grid' displays the following data:

Record_Id	Checkup_Date	Health_Status	Pet_P_Id
401	2025-07-15	Healthy	201
402	2025-07-16	Vaccinated	202
403	2025-07-17	Deformed	203
404	2025-07-18	Minor Injury - Healing	204
405	2025-07-19	Healthy	205
406	2025-07-20	Recovering from surgery	206
407	2025-07-21	Needs dental care	207
408	2025-07-22	Healthy	208
409	2025-07-23	Under observation	209
410	2025-07-24	Vaccinated & Healthy	210

5. Retrieve the records with conditional operators

I. Equal to (=)

SELECT *FROM Staff WHERE Role = 'Veterinarian';

```
1 • SELECT *
2 FROM Staff
3 WHERE Role = 'Manager';
4
```

Result Grid

Staff_Id	Name	Email	Phone	Role
102	Rita Pandey	rita.pandey@center.org	9802222222	Manager
109	Gopal Karki	gopal.karki@center.org	9809999999	Manager

II. Greater Than (>)

SELECT * FROM Medical_Record WHERE Record_Id > 405;

Pet adoption* SQL File 7

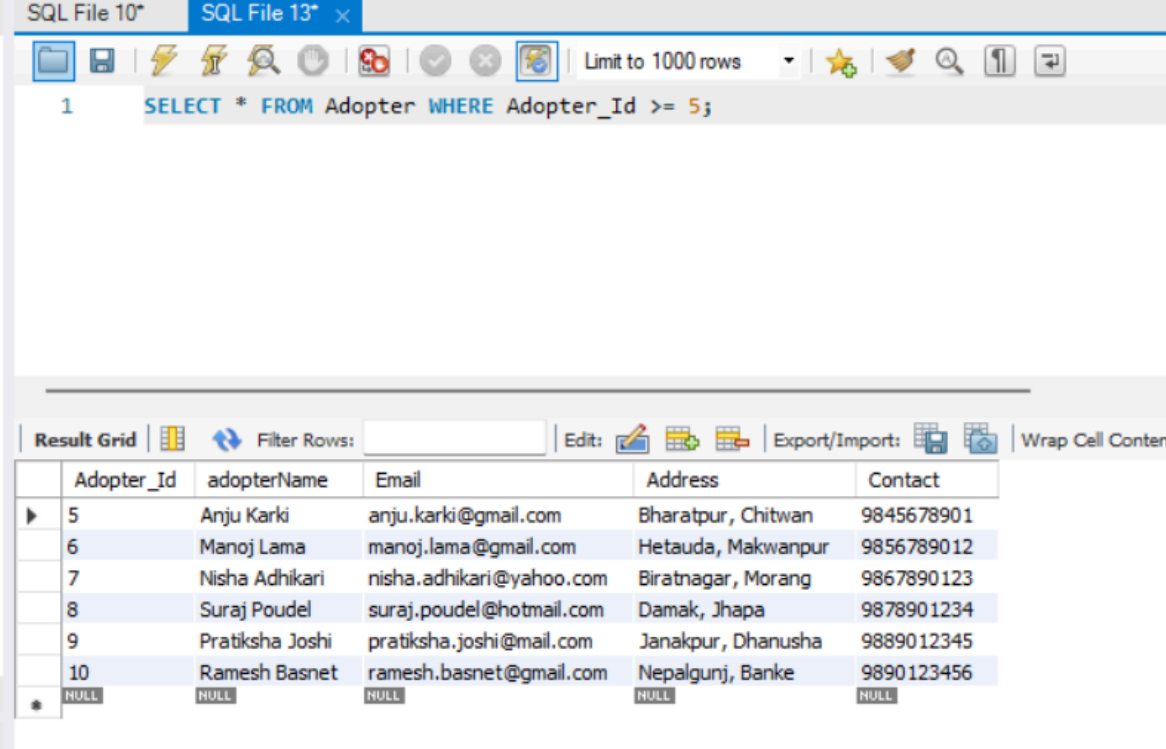
```
1 SELECT * FROM Medical_Record WHERE Record_Id > 405;
```

Result Grid

Record_Id	Checkup_Date	Health_Status	Pet_P_Id
406	2025-07-20	Recovering from surgery	206
407	2025-07-21	Needs dental care	207
408	2025-07-22	Healthy	208
409	2025-07-23	Under observation	209
410	2025-07-24	Vaccinated & Healthy	210

III. Greater Than or Equal To (>=)

```
SELECT * FROM Adopter WHERE Adopter_Id >= 5;
```

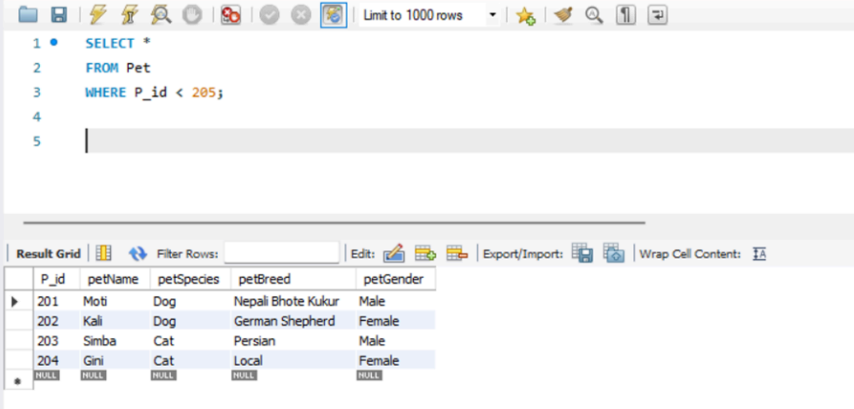


The screenshot shows a SQL IDE with two tabs: 'SQL File 10*' and 'SQL File 13*'. The active tab 'SQL File 13*' contains the query: `1 SELECT * FROM Adopter WHERE Adopter_Id >= 5;`. Below the query editor, the 'Result Grid' displays the results of the query. The grid has columns: Adopter_Id, adopterName, Email, Address, and Contact. The results show 6 rows of data for adopters with IDs 5 through 10, plus a row of NULL values. The interface includes a toolbar with icons for file operations, a 'Limit to 1000 rows' dropdown, and a 'Filter Rows' input field.

	Adopter_Id	adopterName	Email	Address	Contact
▶	5	Anju Karki	anju.karki@gmail.com	Bharatpur, Chitwan	9845678901
	6	Manoj Lama	manoj.lama@gmail.com	Hetauda, Makwanpur	9856789012
	7	Nisha Adhikari	nisha.adhikari@yahoo.com	Biratnagar, Morang	9867890123
	8	Suraj Poudel	suraj.poudel@hotmail.com	Damak, Jhapa	9878901234
	9	Pratiksha Joshi	pratiksha.joshi@mail.com	Janakpur, Dhanusha	9889012345
	10	Ramesh Basnet	ramesh.basnet@gmail.com	Nepalgunj, Banke	9890123456
*	NULL	NULL	NULL	NULL	NULL

IV. Less Than (<)

```
SELECT * FROM Pet WHERE P_id < 205;
```

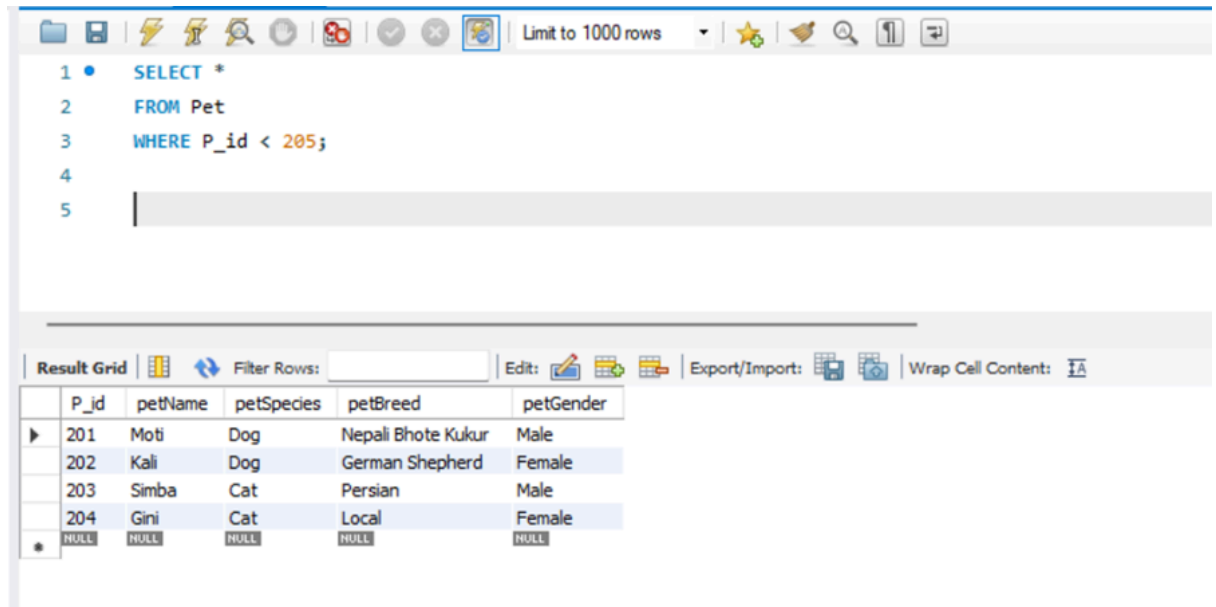


The screenshot shows a SQL IDE with a query editor containing the query: `1 SELECT *
2 FROM Pet
3 WHERE P_id < 205;
4
5`. Below the query editor, the 'Result Grid' displays the results of the query. The grid has columns: P_id, petName, petSpecies, petBreed, and petGender. The results show 4 rows of data for pets with IDs 201 through 204, plus a row of NULL values. The interface includes a toolbar with icons for file operations, a 'Limit to 1000 rows' dropdown, and a 'Filter Rows' input field.

	P_id	petName	petSpecies	petBreed	petGender
▶	201	Moti	Dog	Nepali Bhote Kukur	Male
	202	Kali	Dog	German Shepherd	Female
	203	Simba	Cat	Persian	Male
	204	Gini	Cat	Local	Female
*	NULL	NULL	NULL	NULL	NULL

V. Less Than or Equal To (<=)

SELECT * FROM Adopter WHERE Adopter_Id <= 5;



The screenshot shows a database query editor with a SQL query and its results. The query is:

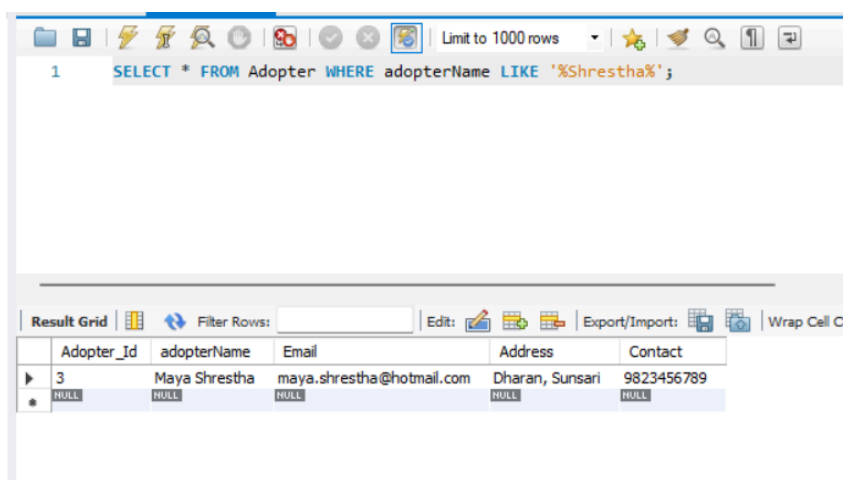
```
1 SELECT *
2 FROM Pet
3 WHERE P_id < 205;
4
5
```

The results are displayed in a table with the following columns: P_id, petName, petSpecies, petBreed, and petGender. The table contains four rows of data:

P_id	petName	petSpecies	petBreed	petGender
201	Moti	Dog	Nepali Bhote Kukur	Male
202	Kali	Dog	German Shepherd	Female
203	Simba	Cat	Persian	Male
204	Gini	Cat	Local	Female

VI. Like

SELECT * FROM Adopter WHERE adopterName LIKE '%Shrestha%';



The screenshot shows a database query editor with a SQL query and its results. The query is:

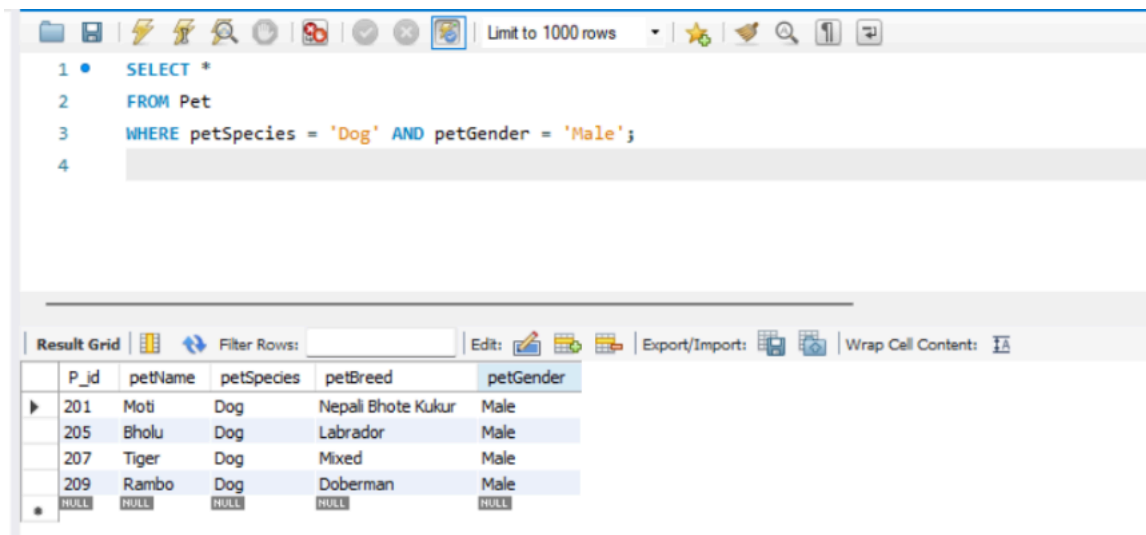
```
1 SELECT * FROM Adopter WHERE adopterName LIKE '%Shrestha%';
```

The results are displayed in a table with the following columns: Adopter_Id, adopterName, Email, Address, and Contact. The table contains one row of data:

Adopter_Id	adopterName	Email	Address	Contact
3	Maya Shrestha	maya.shrestha@hotmail.com	Dharan, Sunsari	9823456789

VII. AND

SELECT * FROM Pet WHERE petSpecies = 'Dog' AND petGender = 'Male';

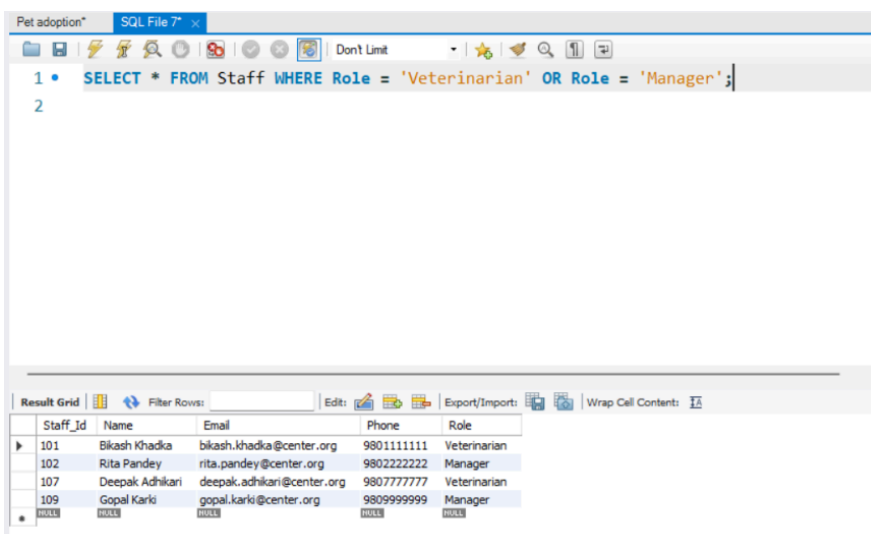


The screenshot shows a SQL IDE window with a query editor and a result grid. The query is: `SELECT * FROM Pet WHERE petSpecies = 'Dog' AND petGender = 'Male';`. The result grid displays the following data:

P_id	petName	petSpecies	petBreed	petGender
201	Moti	Dog	Nepali Bhote Kukur	Male
205	Bholu	Dog	Labrador	Male
207	Tiger	Dog	Mixed	Male
209	Rambo	Dog	Doberman	Male
NULL	NULL	NULL	NULL	NULL

VII. OR

SELECT * FROM Staff WHERE Role = 'Veterinarian' OR Role = 'Manager';



The screenshot shows a SQL IDE window with a query and its results. The query is: `SELECT * FROM Staff WHERE Role = 'Veterinarian' OR Role = 'Manager';`. The result grid displays the following data:

Staff_Id	Name	Email	Phone	Role
101	Bikash Khadka	bikash.khadka@center.org	9801111111	Veterinarian
102	Rita Pandey	rita.pandey@center.org	9802222222	Manager
107	Deepak Adhikari	deepak.adhikari@center.org	9807777777	Veterinarian
109	Gopal Karki	gopal.karki@center.org	9809999999	Manager
NULL	NULL	NULL	NULL	NULL

VIII. NOT

SELECT * FROM Staff WHERE Role != 'Manager';

```
1 • SELECT *
2 FROM Staff
3 WHERE Role != 'Manager';
4
```

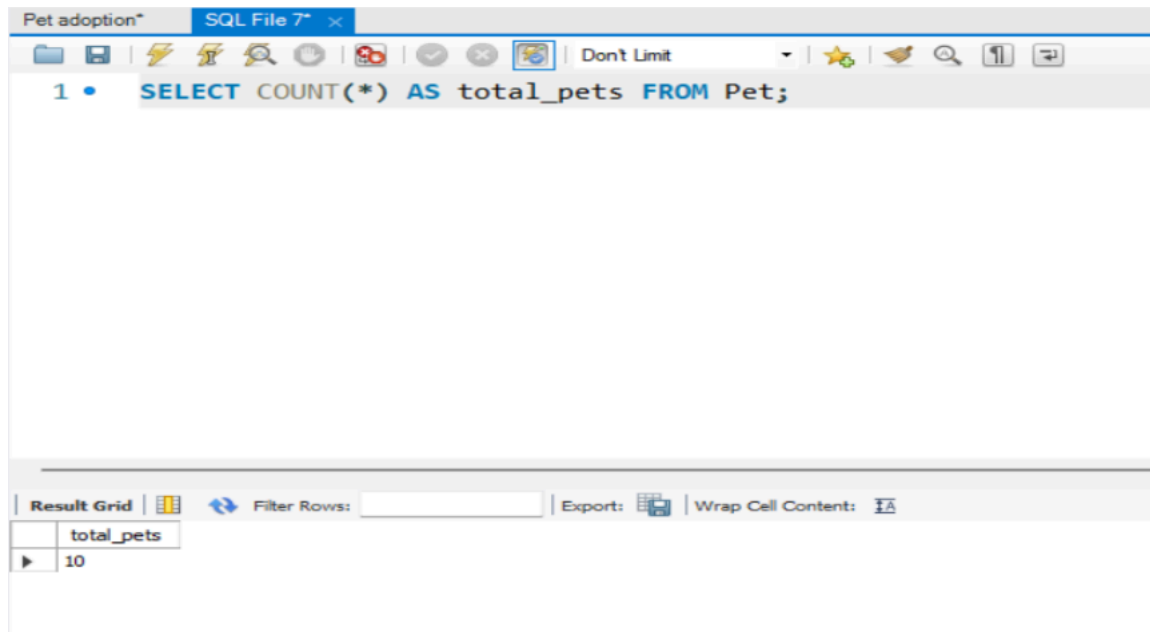
Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
Staff_Id	Name	Email	Phone	Role
101	Bikash Khadka	bikash.khadka@center.org	9801111111	Veterinarian
103	Suman Giri	suman.giri@center.org	9803333333	Caretaker
104	Laxmi Bista	laxmi.bista@center.org	9804444444	Receptionist
105	Kamal Rana	kamal.rana@center.org	9805555555	Cleaner
106	Sujata Sharma	sujata.sharma@center.org	9806666666	Caretaker
107	Deepak Adhikari	deepak.adhikari@center.org	9807777777	Veterinarian
108	Asmita Neupane	asmita.neupane@center.org	9808888888	Volunteer
110	Pooja Regmi	pooja.regmi@center.org	9810000000	Receptionist
NULL	NULL	NULL	NULL	NULL

6. Use aggregate functions

I. COUNT

The COUNT function returns the number of rows that match a specified condition.

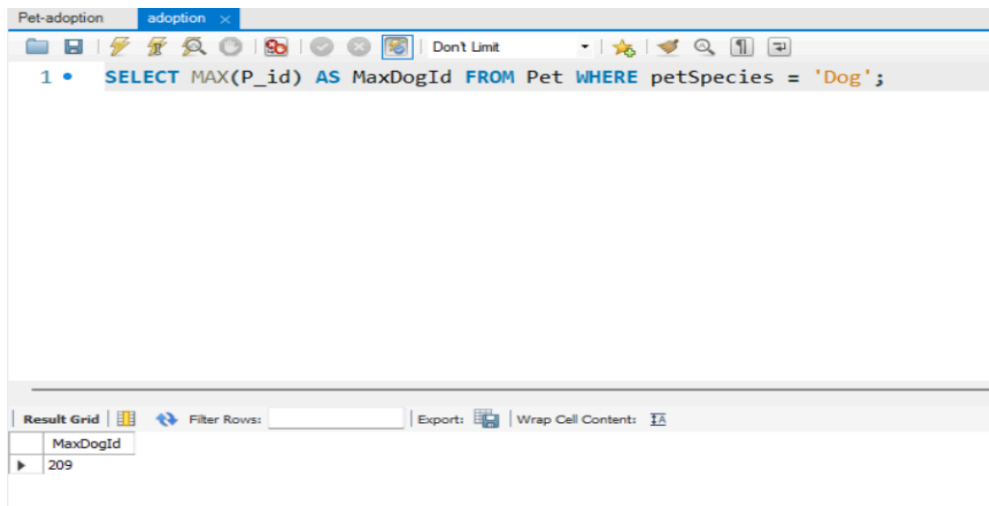
Example: Count the total number of pets in the Pet table:



II. MAX

The MAX function returns the highest value in a specified column.

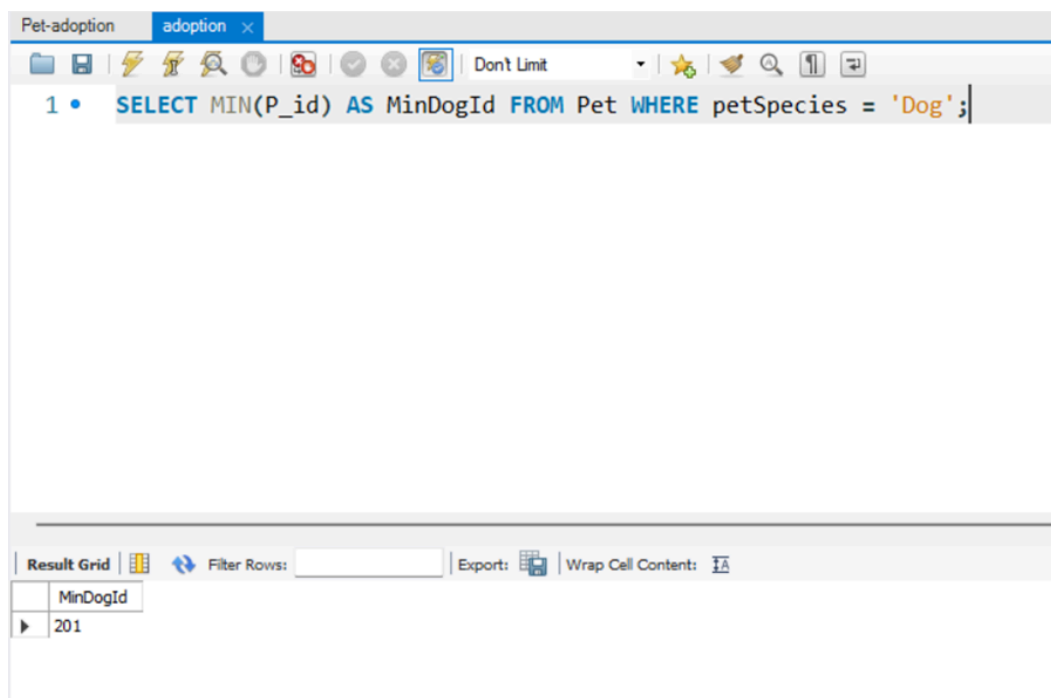
Example: Find the highest Pet ID assigned to a Dog in the Pet table?



III. MIN

The MIN function returns the lowest value in a specified column.

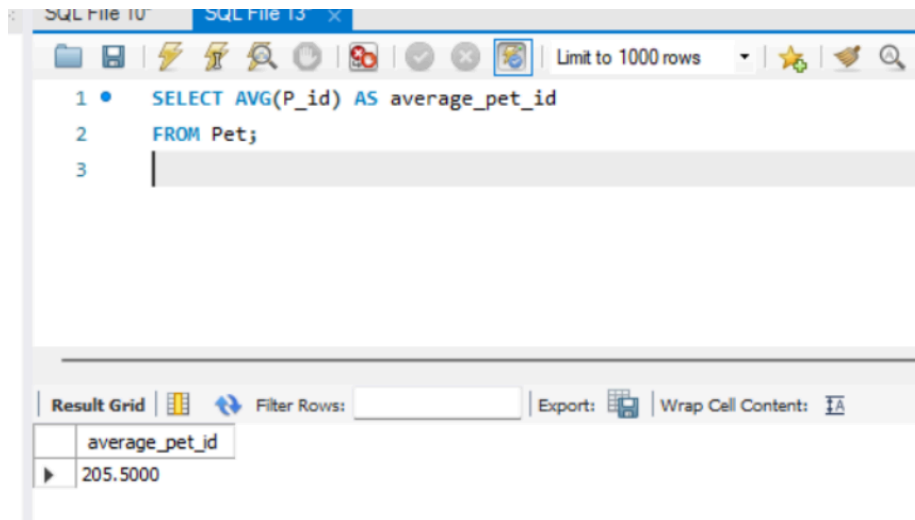
Example: Find the Lowest Pet ID assigned to a Dog in the Pet table?



IV.AVG

The AVG function returns the average value of a numeric column.

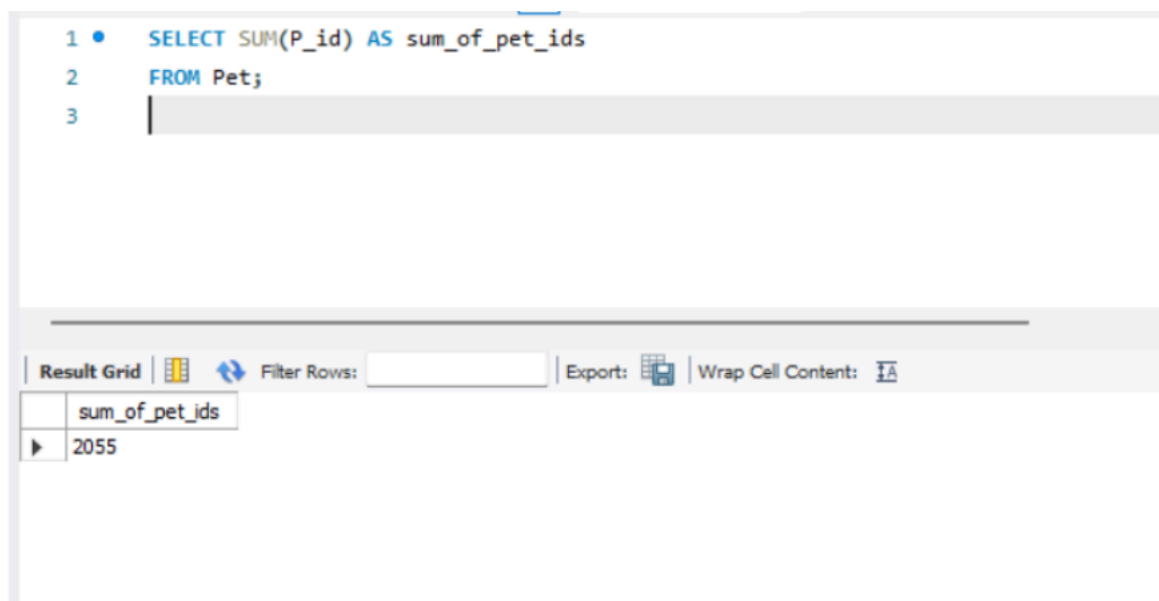
Example: Find the average pet ID



V. SUM

The SUM function returns the total sum of a numeric column.

Example: Find the sum of all Pet IDs



7. Use joins

1.Inner

```
SELECT A.adopterName, P.petName FROM Adoption AD JOIN Adopter A ON  
AD.Adopter_Id = A.Adopter_Id
```

```
INNER JOIN Pet P ON AD.Pet_id = P.P_id;
```

```
1 • SELECT A.adopterName, P.petName  
2 FROM Adoption AD  
3 JOIN Adopter A ON AD.Adopter_Id = A.Adopter_Id  
4 INNER JOIN Pet P ON AD.Pet_id = P.P_id;  
5
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	adopterName	petName			
▶	Sita Thapa	Moti			
	Rajesh Gurung	Kali			
	Maya Shrestha	Simba			
	Dipesh Bhattarai	Gini			
	Anju Karki	Bholu			
	Manoj Lama	Luna			
	Nisha Adhikari	Tiger			
	Suraj Poudel	Sweetie			
	Pratiksha Joshi	Rambo			
	Ramesh Basnet	Pilu			

II. LEFT JOIN

```
SELECT P.petName, AD.Date AS AdoptionDate FROM Pet P LEFT JOIN Adoption AD  
ON P.P_id = AD.Pet_id;
```

The screenshot shows a database query editor with a toolbar at the top. The query text is as follows:

```
1 • SELECT P.petName, AD.Date AS AdoptionDate
2 FROM Pet P
3 LEFT JOIN Adoption AD ON P.P_id = AD.Pet_id;
4
```

Below the query editor, the 'Result Grid' is displayed, showing the results of the query. The grid has two columns: 'petName' and 'AdoptionDate'. The results are as follows:

petName	AdoptionDate
Moti	2025-08-01
Kali	2025-08-02
Simba	2025-08-03
Gini	2025-08-04
Bholu	2025-08-05
Luna	2025-08-06
Tiger	2025-08-07
Sweetie	2025-08-08
Rambo	2025-08-09
Pilu	2025-08-10

III. RIGHT JOIN

SELECT AD.Adoption_Id, P.petName FROM Adoption AD RIGHT JOIN Pet P ON
AD.Pet_id = P.P_id;



Limit to 1000 rows

1 •

```
SELECT AD.Adoption_Id, P.petName
FROM Adoption AD
RIGHT JOIN Pet P ON AD.Pet_id = P.P_id;
```

4

Result Grid



 Filter Rows:

Export: 

Wrap C

	Adoption_Id	petName
▶	1	Moti
	2	Kali
	3	Simba
	4	Gini
	5	Bholu
	6	Luna
	7	Tiger
	8	Sweety
	9	Rambo
	10	Pilu

8. Create views

CREATE VIEW AdoptionDetails AS

SELECT

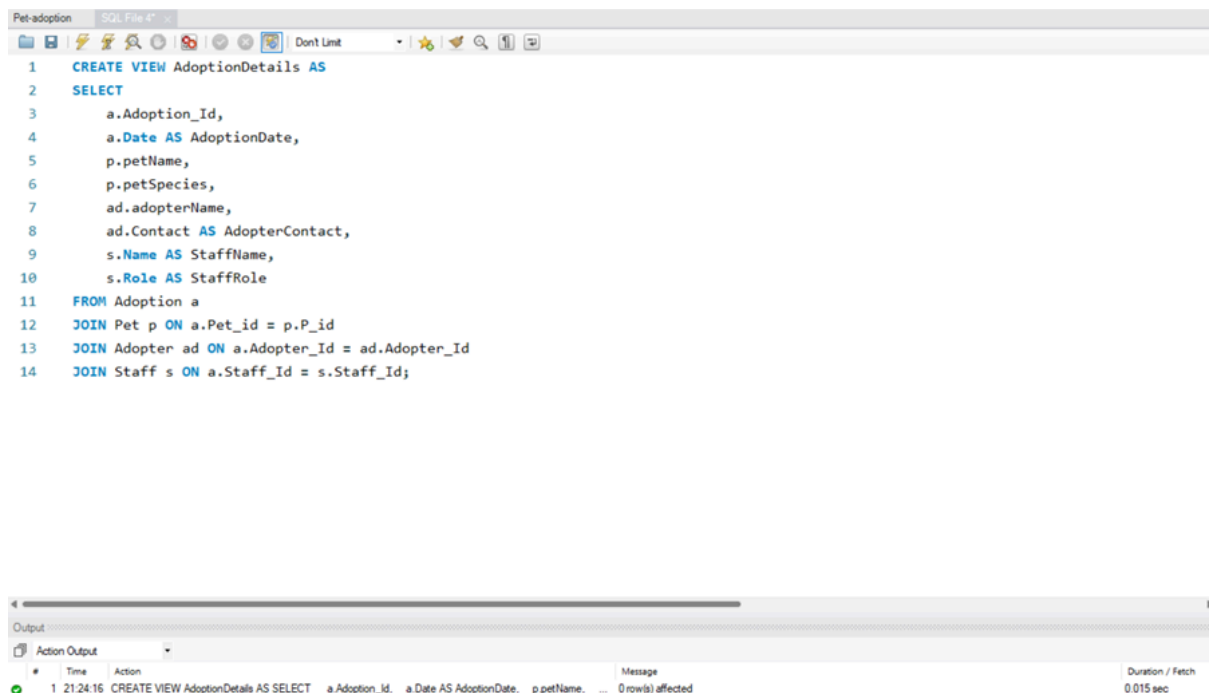
```
a.Adoption_Id,  
a.Date AS AdoptionDate,  
p.petName,  
p.petSpecies,  
ad.adopterName,  
ad.Contact AS AdopterContact,  
s.Name AS StaffName,  
s.Role AS StaffRole
```

FROM Adoption a

JOIN Pet p ON a.Pet_id = p.P_id

JOIN Adopter ad ON a.Adopter_Id = ad.Adopter_Id

JOIN Staff s ON a.Staff_Id = s.Staff_Id;

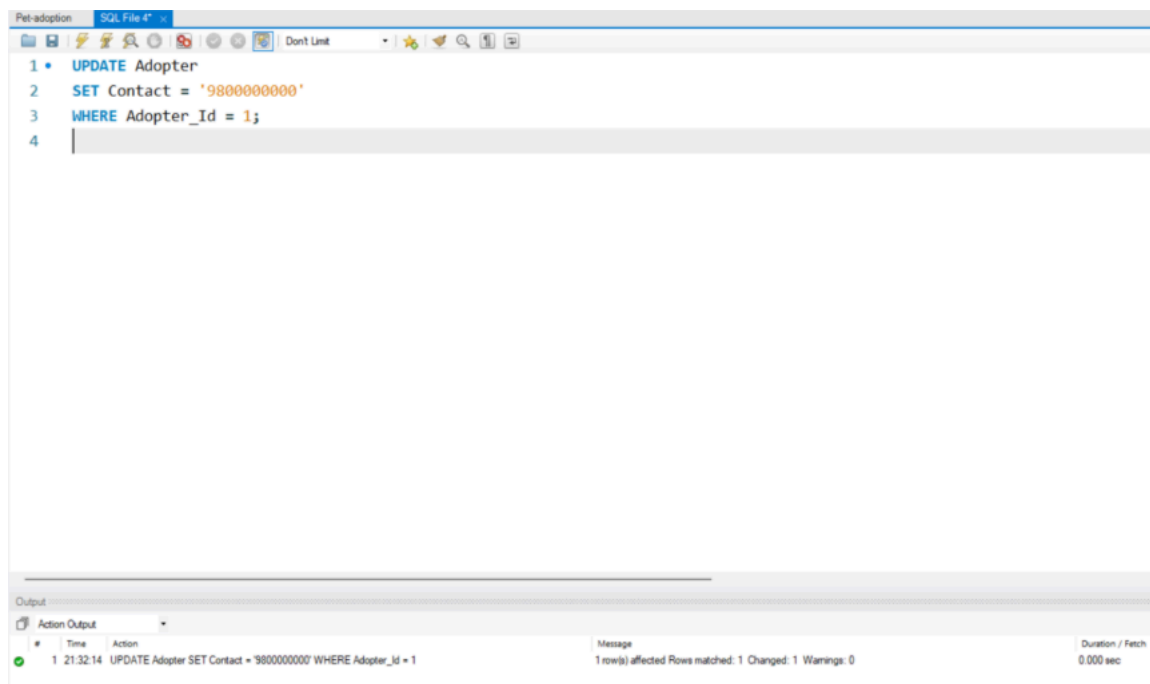


9. Update records

UPDATE Adopter

SET Contact = '9800000000'

WHERE Adopter_Id = 1;



10. Delete records

```
DELETE FROM Medical_Record
```

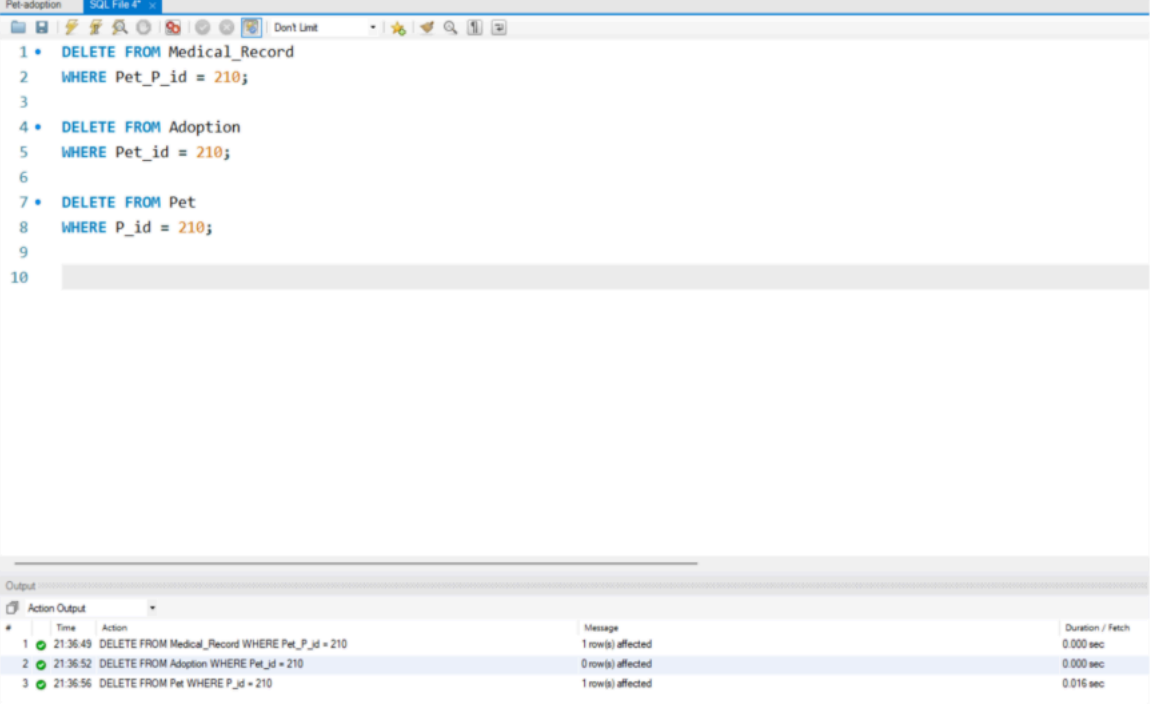
```
WHERE Pet_P_id = 210;
```

```
DELETE FROM Adoption
```

```
WHERE Pet_id = 210;
```

```
DELETE FROM Pet
```

```
WHERE P_id = 210;
```

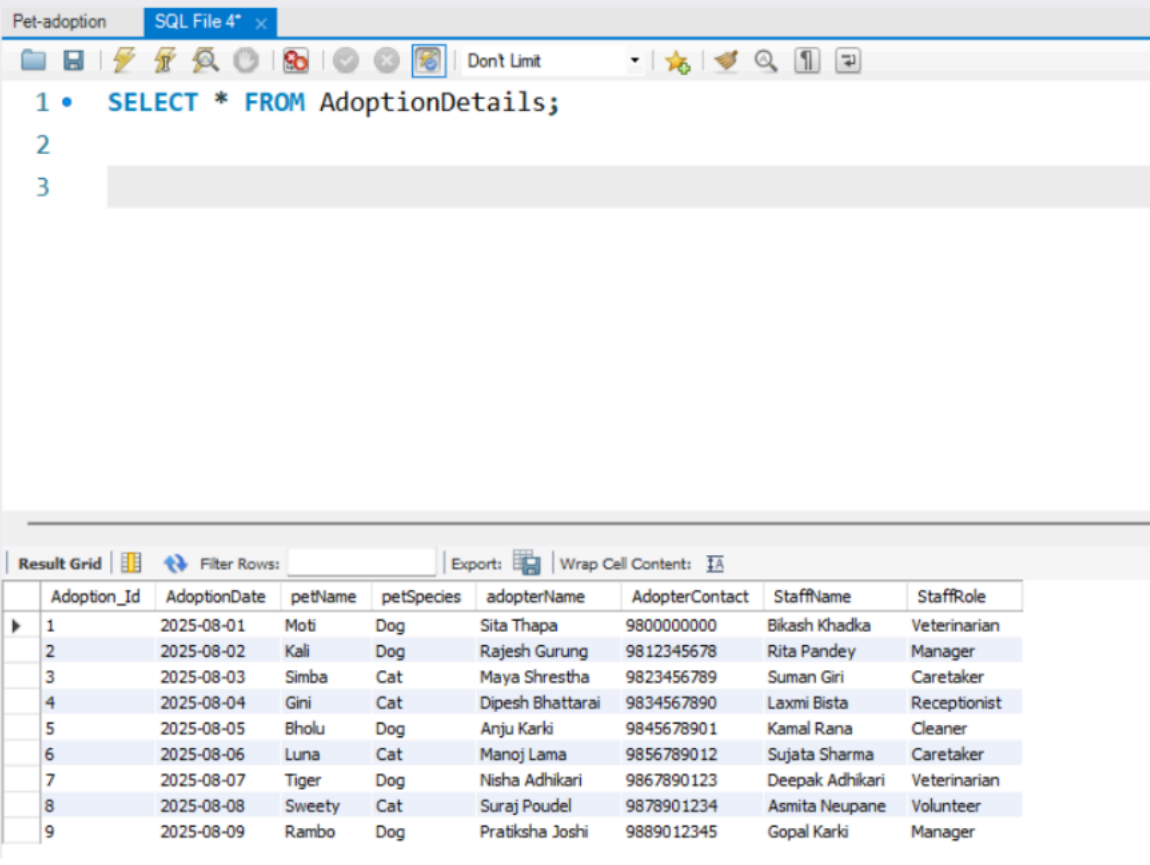


The screenshot shows a SQL script in a window titled "Pet-adoption SQL File 4". The script contains three DELETE statements. Below the script, the "Output" window shows the execution results for each statement.

#	Time	Action	Message	Duration / Fetch
1	21:36:49	DELETE FROM Medical_Record WHERE Pet_P_id = 210	1 row(s) affected	0.000 sec
2	21:36:52	DELETE FROM Adoption WHERE Pet_id = 210	0 row(s) affected	0.000 sec
3	21:36:56	DELETE FROM Pet WHERE P_id = 210	1 row(s) affected	0.016 sec

11. Check views if updated data is reflected

```
SELECT * FROM AdoptionDetails;
```



The screenshot shows a SQL IDE window titled "Pet-adoption" with a tab for "SQL File 4* x". The query editor contains the SQL statement `SELECT * FROM AdoptionDetails;`. Below the editor, the "Result Grid" displays the query results in a table format. The table has 9 rows and 8 columns: Adoption_Id, AdoptionDate, petName, petSpecies, adopterName, AdopterContact, StaffName, and StaffRole. The data is as follows:

Adoption_Id	AdoptionDate	petName	petSpecies	adopterName	AdopterContact	StaffName	StaffRole
1	2025-08-01	Moti	Dog	Sita Thapa	9800000000	Bikash Khadka	Veterinarian
2	2025-08-02	Kali	Dog	Rajesh Gurung	9812345678	Rita Pandey	Manager
3	2025-08-03	Simba	Cat	Maya Shrestha	9823456789	Suman Giri	Caretaker
4	2025-08-04	Gini	Cat	Dipesh Bhattarai	9834567890	Laxmi Bista	Receptionist
5	2025-08-05	Bholu	Dog	Anju Karki	9845678901	Kamal Rana	Cleaner
6	2025-08-06	Luna	Cat	Manoj Lama	9856789012	Sujata Sharma	Caretaker
7	2025-08-07	Tiger	Dog	Nisha Adhikari	9867890123	Deepak Adhikari	Veterinarian
8	2025-08-08	Sweetie	Cat	Suraj Poudel	9878901234	Asmita Neupane	Volunteer
9	2025-08-09	Rambo	Dog	Pratiksha Joshi	9889012345	Gopal Karki	Manager

12. Alter views

ALTER VIEW AdoptionDetails AS

SELECT

 a.Adoption_Id,
 a.Date AS AdoptionDate,
 p.petName,
 p.petSpecies,
 ad.adopterName,
 ad.Email AS AdopterEmail,
 ad.Contact AS AdopterContact,
 s.Name AS StaffName,
 s.Role AS StaffRole

FROM Adoption a

JOIN Pet p ON a.Pet_id = p.P_id

JOIN Adopter ad ON a.Adopter_Id = ad.Adopter_Id

JOIN Staff s ON a.Staff_Id = s.Staff_Id;

```

1  ALTER VIEW AdoptionDetails AS
2  SELECT
3      a.Adoption_Id,
4      a.Date AS AdoptionDate,
5      p.petName,
6      p.petSpecies,
7      ad.adopterName,
8      ad.Email AS AdopterEmail,
9      ad.Contact AS AdopterContact,
10     s.Name AS StaffName,
11     s.Role AS StaffRole
12 FROM Adoption a
13 JOIN Pet p ON a.Pet_id = p.P_id
14 JOIN Adopter ad ON a.Adopter_Id = ad.Adopter_Id
15 JOIN Staff s ON a.Staff_Id = s.Staff_Id;
16

```

13. Drop views

DROP VIEW AdoptionDetails;

